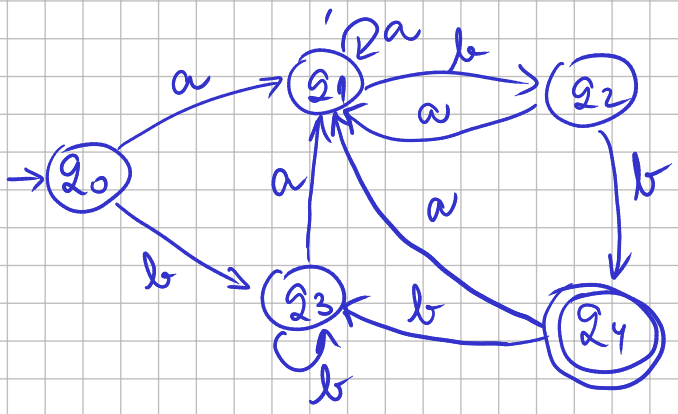
# Limbaje Formale și Automate
## Seminarul 7

### Algoritmul de minimizare

1. Determinați AFD echivalent minim pentru:




Pornim de la 2 partiții

$P = \{q_0, q_1, q_2, q_3\}$ - stări nefinale $\quad F = \{q_4\}$ - stare finală

$$\left.\begin{array}{l} m(q_0, a) = q_1 \in P \\ m(q_1, a) = q_1 \in P \\ m(q_2, a) = q_1 \in P \\ m(q_3, a) = q_1 \in P \end{array}\right\} \Rightarrow q_0 \equiv q_1 \equiv q_2 \equiv q_3 \text{ pt. } a$$

$$\left.\begin{array}{l} m(q_0, b) = q_3 \in P \\ m(q_1, b) = q_2 \in P \\ m(q_2, b) = q_4 \in F \\ m(q_3, b) = q_3 \in P \end{array}\right\} \Rightarrow q_0 \equiv q_1 \equiv q_3 \text{ pt. } b$$

$$q_0 \not\equiv q_2,\ q_1 \not\equiv q_2,\ q_3 \not\equiv q_2 \Rightarrow$$

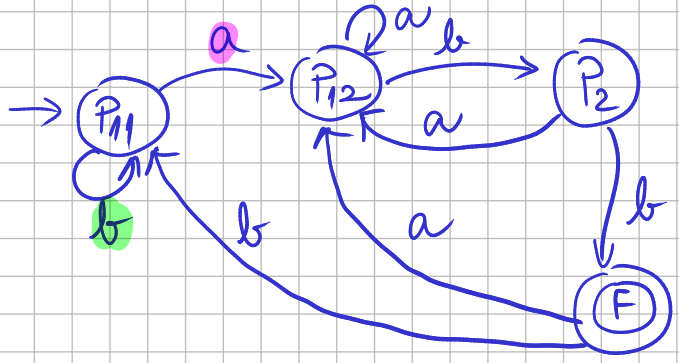
$\Rightarrow$ spargem partiția $P$ în $P_1 = \{q_0, q_1, q_3\}$, $P_2 = \{q_2\}$

$$\left.\begin{array}{l} m(q_0, a) = q_1 \in P_1 \\ m(q_1, a) = q_1 \in P_1 \\ m(q_3, a) = q_1 \in P_1 \end{array}\right.$$

$$\left.\begin{array}{l} m(q_0, b) = q_3 \in P_1 \\ m(q_1, b) = q_2 \in P_2 \\ m(q_3, b) = q_3 \in P_1 \end{array}\right\} \Rightarrow \text{spargem partiția } P_1 \text{ în } P_{11} = \{q_0, q_3\} \text{ și } P_{12} = \{q_1\}$$

$m(q_0, a) = q_1 \in P_{12}$ $\quad m(q_0, b) = q_3 \in P_{11}$

$m(q_3, a) = q_1 \in P_{12}$ $\quad m(q_3, b) = q_3 \in P_{11}$ $\Big\} \Rightarrow q_0 \equiv q_3$




AFD minim

2. Aplicați alg. de minimizare pt. următorul AFD:

| | a | b |
|---|---|---|
| → $q_0$ | $q_1$ | $q_3$ |
| $q_1$ | $q_1$ | $q_2$ |
| $q_2$ | $q_2$ | $q_1$ |
| $q_3$ | $q_5$ | $q_3$ |
| $q_4$ | $q_3$ | $q_5$ |
| $q_5$ | $q_5$ | $q_5$ |

$P = \{q_0, q_1, q_2, q_4\}$

$F = \{q_3, q_5\}$

Verificăm pt. P

$m(q_0, a) = q_1 \in P$

$m(q_1, a) = q_1 \in P$

$m(q_2, a) = q_2 \in P$

$m(q_4, a) = q_3 \in F$

$\Big\} \Rightarrow q_0 \equiv q_1 \equiv q_2$ pt. a trebuie să spargem P în $P_1 = \{q_0, q_1, q_2\}$

$P_2 = \{q_4\}$

Verificăm pt. $P_1$

$m(q_0, a) = q_1 \in P_1$ $\quad m(q_0, b) = q_3 \in F$

$m(q_1, a) = q_1 \in P_1$ $\quad m(q_1, b) = q_2 \in P_1$

$m(q_2, a) = q_2 \in P_1$ $\quad m(q_2, b) = q_1 \in P_1$

$\Big\} \Rightarrow$ spargem $P_1$ în

$P_{11} = \{q_0\}$

$P_{12} = \{q_1, q_2\}$

Verificăm pt. $P_{12}$

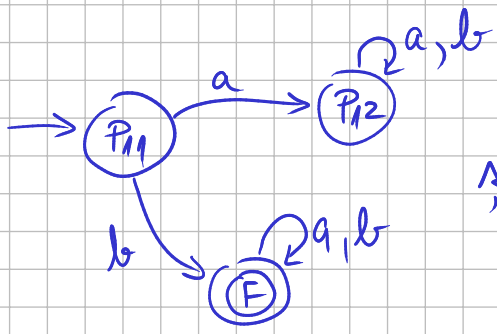
$m(q_1, a) = q_1 \in P_{12}$ $\quad m(q_1, b) = q_2 \in P_{12}$

$m(q_2, a) = q_2 \in P_{12}$ $\quad m(q_2, b) = q_1 \in P_{12}$ $\Big\} \Rightarrow q_1 \equiv q_2$

Verificăm pt. F

$m(Q_3, a) = Q_5 \in F$ $\quad m(Q_3, b) = Q_3 \in F$

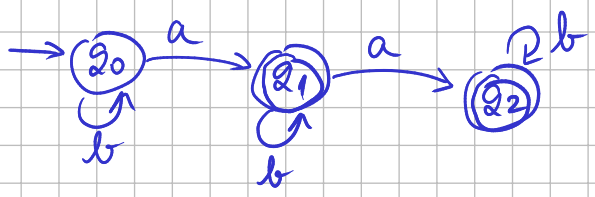
$m(Q_5, a) = Q_5 \in F$ $\quad m(Q_5, b) = Q_5 \in F$ $\Big\} \Rightarrow Q_3 \equiv Q_5$




șiruri care încep cu b

3. Propuneți un AFD care acceptă:

$L_1 = \{ w \in \{a, b\}^* \mid \#_a(w) \text{ este 1 sau 2} \}$



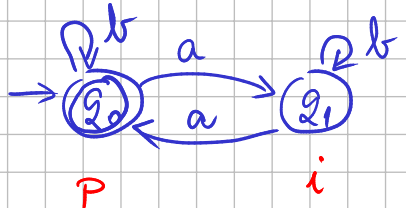

$L_2 = \{ w \in \{a, b\}^* \mid \#_a(w) \bmod 2 = 0 \text{ și } \#_b(w) \bmod 3 = 1 \}$

$L_2 = L_2' \cap L_2''$

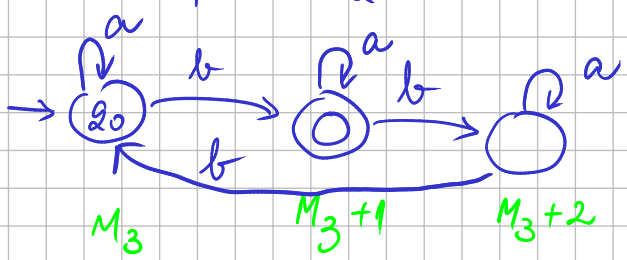
$L_2' = \{ w \in \{a, b\}^* \mid \#_a(w) \bmod 2 = 0 \}$

$L_2'' = \{ w \in \{a, b\}^* \mid \#_b(w) \bmod 3 = 1 \}$

AFD pt. $L_2'$




AFD pt. $L_2''$




Intersecția a două automate

Fie $M_1$ un AFD care acceptă $L_1$ : $M_1 = (Q_1, \Sigma, m_1, q_1, F_1)$

Fie $M_2$ un AFD care acceptă $L_2$ : $M_2 = (Q_2, \Sigma, m_2, q_2, F_2)$

Atunci putem obține un AFD $M$ care acceptă $L_1 \cap L_2$

unde $M = (Q, \Sigma, m, q_0, F)$

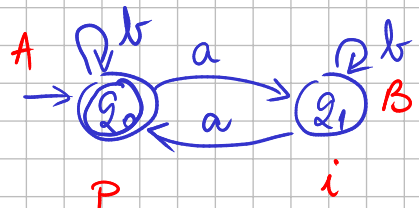
$Q = \{ (r_1, r_2) \mid r_1 \in Q_1 \text{ și } r_2 \in Q_2 \}$ – produs cartezian

$m((r_1, r_2), a) = (m(r_1, a), m(r_2, a)), \forall (r_1, r_2) \in Q \text{ și } a \in \Sigma$

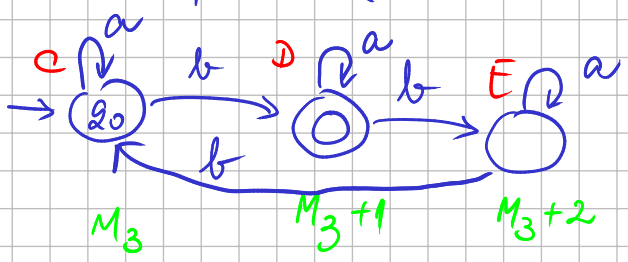
$q_0 = (q_1, q_2)$

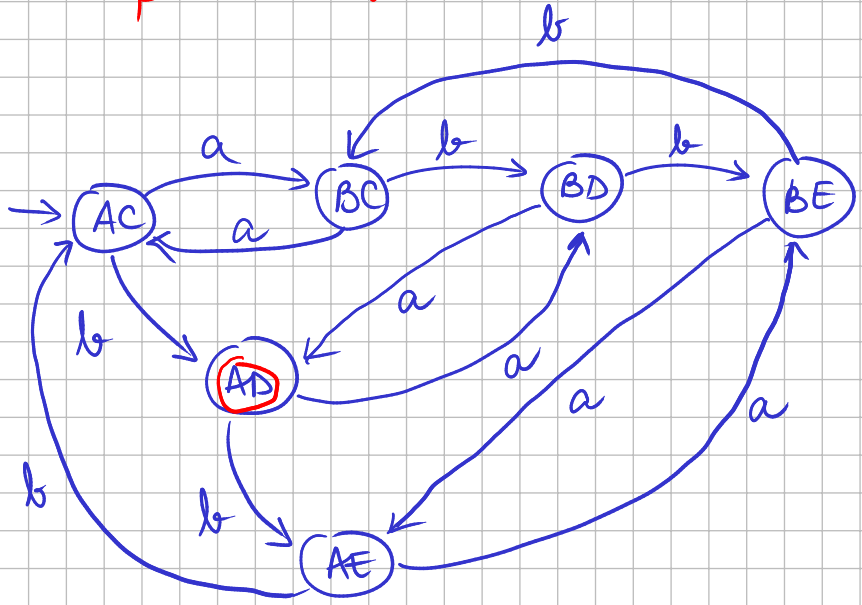
$F = \{ (r_1, r_2) \mid r_1 \in F_1 \text{ și } r_2 \in F_2 \}$

AFD pt. $L_2'$



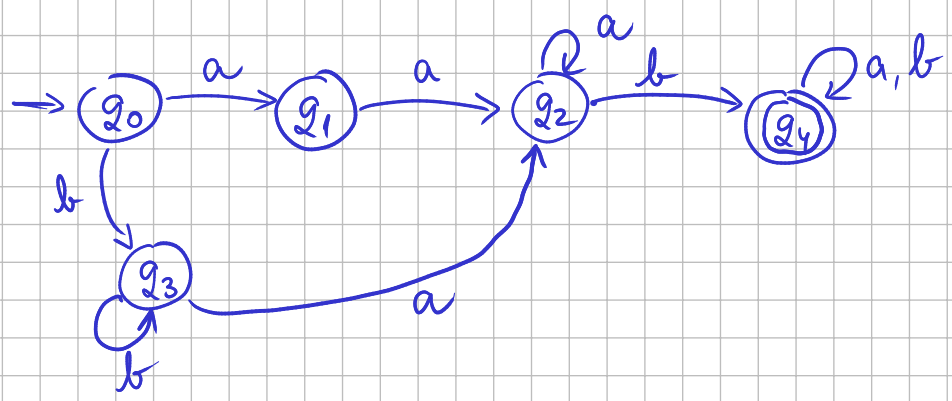

AFD pt. $L_2''$



Scrieți un AFD care acceptă

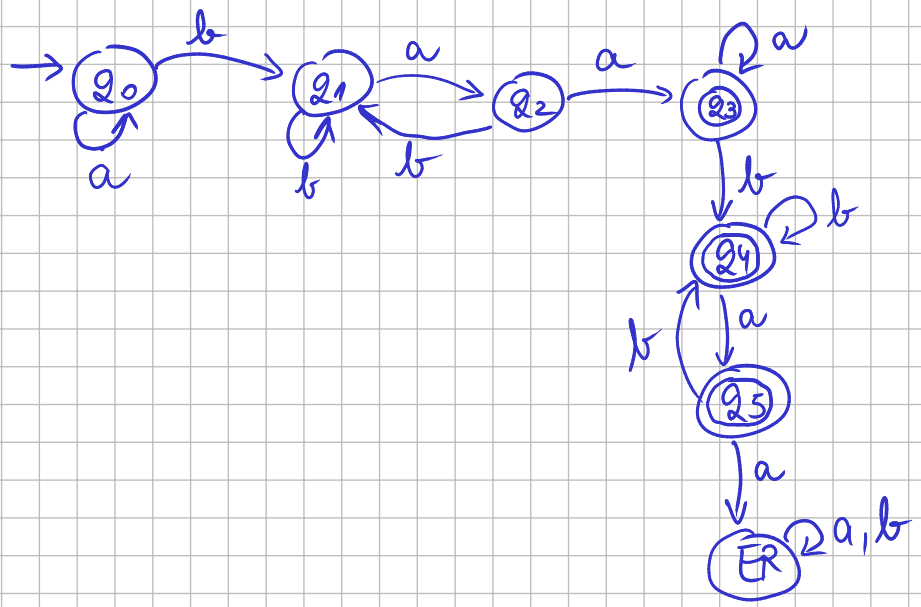
$L = \{ w \in \{a,b\}^* \mid w$ conține $ab$, dar nu începe cu $ab\}$




baab

Propuneți un AFD care acceptă

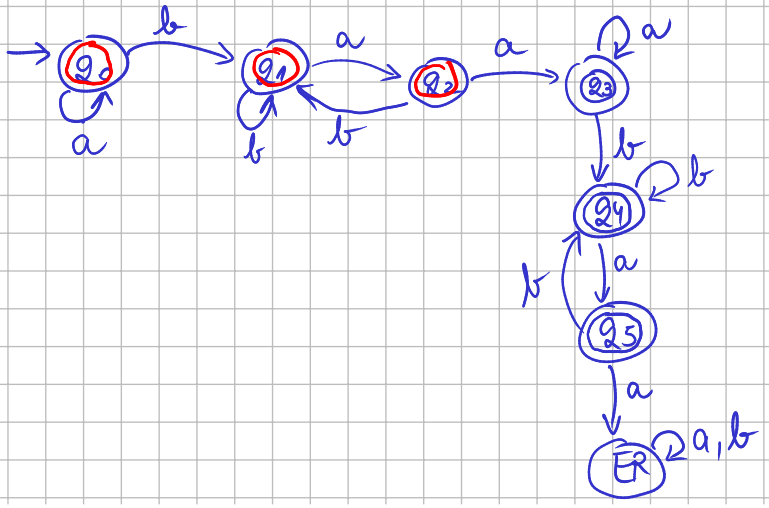
$L_1 = \{ w \in \{a,b\}^* \mid w$ conține $baa$ o singură dată$\}$




$baab \in L_1$

$baaba \in L_1$

$L_2 = \{ w \in \{a,b\}^* \mid w$ conține $baa$ cel mult o dată$\}$

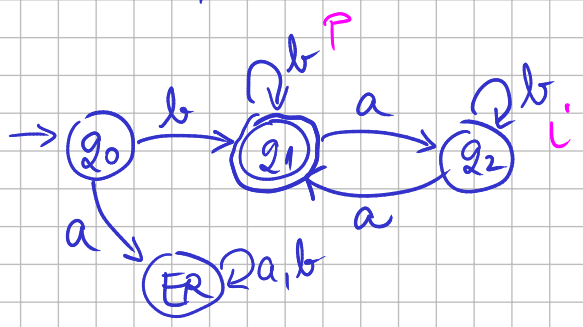


$L_3 = \{ w \in \{a,b\}^* \mid w$ începe cu b și $\#a(w)\%2 = 0$
sau
$w$ începe cu a$\}$

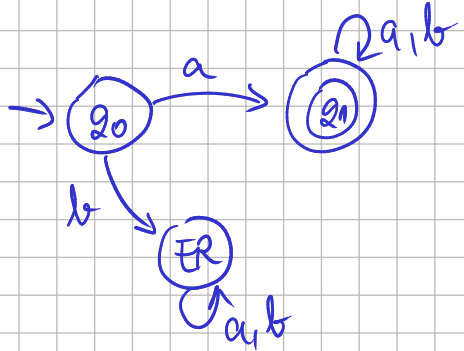
$L_3 = (L_3' \cap L_3'') \cup L_3'''$

$L_3 = \{ w \in \{a,b\}^* \mid w$ începe cu b ($L_3'$) și $\#a(w)\%2 = 0 \} \cup$
$\{ w \in \{a,b\}^* \mid w$ începe cu a$\}$ ($L_3''$)

AFD pt. $L'$




AFD pt. $L''$

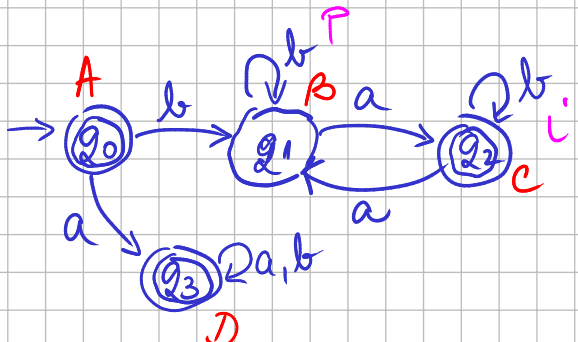



$L_1 \cup L_2 = \overline{\overline{L_1} \cap \overline{L_2}}$

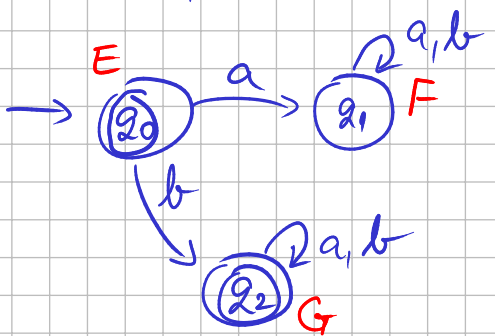
Construcția AFD pt. reuniune

1. Determinăm AFD pt. $\overline{L_1}$
2. Determinăm AFD pt. $\overline{L_2}$
3. Realizăm intersecția automatelor pt. $\overline{L_1}$ și $\overline{L_2}$
4. Determinăm complementul AFD de la pasul 3.

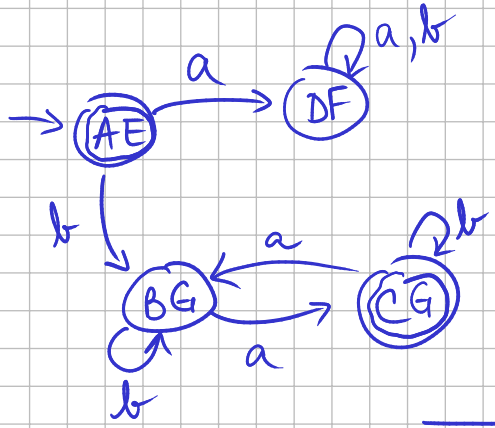
1. AFD pt. $\overline{L_1}$




2. AFD pt $\overline{L_2}$



3. AFD pt. $\overline{L_1} \cap \overline{L_2}$




4. AFD pt $\overline{\overline{L_1} \cap \overline{L_2}}$

